Final Project Progress Report

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1 Project Name

Testing-Oriented Agent System

2 Overview

Our project's primary objective is to develop and deploy a testing-oriented LLM agent system for software testing at the repository level. As previous research on LLM-based agents has also indicated, we believe that software testing is not a one-step activity but rather requires numerous rounds of reasoning and code development.

An efficient testing agent should be able to search through a repository, comprehend what has to be tested, organize the testing procedures, create and execute numerous tests, assess failures, and gradually improve the outcomes. In this case, it should also be able to generate an intelligible final report. Our main objective is to create a more organized agent system that can facilitate the entire testing process rather than merely performing discrete, small-scale coding tasks.

We intend to test our method on small to medium-sized open-source Python repositories in order to assess it. Our goal is to determine whether a customized testing agent can indeed outperform more straightforward LLM techniques in terms of efficiency and dependability. We pay special attention to activities including preparation, execution, enhancing test coverage, and failure analysis. Our ultimate goal is to produce a functional prototype and investigate ways to enhance LLM agents for actual software testing situations, where workflow control and reproducibility are crucial.

3 Research Questions

RQ1: How can the planning, creation, execution, and iterative improvement of software testing activities at the repository level be supported by an LLM agent system?

RQ2: How much can a testing-oriented agent system increase the efficiency and dependability of software testing when compared to more straightforward LLM-based techniques (such as direct test generation or simple ReAct loops)?

4 Value to User Community

Software developers, QA engineers, and researchers interested in LLM-based agent systems would be the project's primary users.

The approach might save developers and testers the time and effort required to comprehend repositories, plan tests, execute them, and troubleshoot errors—particularly for crucial but repetitive testing chores.

This project can provide academics with a useful illustration of how to create a testing-focused agent system that operates at the repository level rather than only producing code in a single step. The most significant benefit of this work, in our opinion, is the provision of a more comprehensive and practical testing workflow, which is now not fully supported by coding assistants.

In order for others to examine, replicate, and potentially expand upon our work, we also want to publish and push our work—including code, documentation, and examples—to GitHub.

5 Demo

We will describe our project as a testing-oriented LLM agent system intended for repository-level testing in the one-minute elevator pitch. We will clarify that although current tools are capable of producing codes or test cases, they typically do not provide adequate assistance for the entire testing process.

In an effort to address this, our solution gives the agent the ability to examine a repository, organize testing procedures, create and run tests, assess errors, and generate final results. We want to stress that testing is a process that involves reasoning, iterations, and the use of tools rather than being a one-step generation problem.

We intend to provide a functional prototype on a modest or midsize Python repository during the five-minute demonstration. The entire procedure will be demonstrated, starting with a repository and testing objective as input, assessing the structure, designing, creating, and executing tests, and finally summarizing outcomes such as coverage or failures.

If time permits, we will also demonstrate an instance of iterative improvement, in which the agent modifies its testing approach in response to prior outcomes. The objective is to show that our system is capable of handling end-to-end testing rather than only generating discrete code.

6 Delivery

We plan to deliver our project through a public GitHub repository:

https://github.com/YihAn011/Testing-Oriented-Agent-System.git

The repository will contain our documentation, source code, and any other resources required to comprehend and replicate the system. Additionally, we will turn in our presentation slides and final report. We shall properly cite any external datasets, models, or code that we utilize.